Universal Inquiry

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Metadefinition 1 (Formula)**.** If a, b are variables, then $a \in b$ is a formula. If p is a formula, then $\neg p$ is a formula. If p, q are formulas, then $p \wedge q$ is a formula. If x is a variable, and p is a formula, then $\exists x\,(p)$ is a formula.

Metadefinition 2. If $p(x)$ is a formula, and y is a variable, then $p(y)$ is $p(x)$ with each occurrence of x replaced with y.

Notation 1 (Disjunction)**.**

$$p \vee q \equiv \neg(\neg p \wedge \neg q)$$

Notation 2 (Implication)**.**

$$p \Rightarrow q \equiv \neg(p \wedge \neg q)$$

Notation 3 (Equivalence)**.**

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

Notation 4 (Exclusive Disjunction)**.**

$$p \oplus q \equiv \neg(p \Leftrightarrow q)$$

Notation 5 (Bounded Existential Quantification)**.**

$$\exists x \in A\,(p) \equiv \exists x\,(x \in A \wedge p)$$

Notation 6 (Typed Existential Quantification)**.**

$$\exists_p x\,(q) \equiv \exists x\,(p(x) \wedge q)$$

Notation 7 (Universal Quantification)**.**

$$\forall x\,(p) \equiv \neg\exists x\,(\neg p)$$

Notation 8 (Bounded Universal Quantification).

$$\forall x \in A\,(p) \equiv \forall x\,(x \in A \Rightarrow p)$$

Notation 9 (Typed Universal Quantification).

$$\forall_p x\,(q) \equiv \forall x\,(p(x) \Rightarrow q)$$

Notation 10 (Equality).

$$X = Y \equiv \forall a\,(a \in X \Leftrightarrow a \in Y) \wedge \forall B\,(X \in B \Leftrightarrow Y \in B)$$

Notation 11 (Unique Quantification).

$$\exists! x\,(p(x)) \equiv \exists x\,(p(x) \wedge \forall y\,(p(y) \Rightarrow y = x))$$

Axiom 1 (Empty Set).

$$\exists n\,(\nexists x\,(x \in n))$$

Definition 1 (Empty Set).

$$n = \varnothing \Leftrightarrow \nexists x\,(x \in n)$$

Axiom 2 (Extensionality).

$$\forall A,\, \forall B\,(\forall x\,(x \in A \Leftrightarrow x \in B) \Rightarrow A = B)$$

Axiom 3 (Pairing).

$$\forall a,\, \forall b,\, \exists C,\, \forall c\,(c \in C \Leftrightarrow (c = a \vee c = b))$$

Definition 2 (Couple).

$$C = \{a, b\} \Leftrightarrow \forall c\,(c \in C \Leftrightarrow (c = a \vee c = b))$$

Notation 12 (Singleton).

$$\{a\} = \{a, a\}$$

Axiom 4 (Union).

$$\forall B,\, \exists U,\, \forall c\,(c \in U \Leftrightarrow \exists P \in B\,(c \in P))$$

Definition 3 (Union).

$$U = \bigcup B \Leftrightarrow \forall c\,(c \in U \Leftrightarrow \exists P \in B\,(c \in P))$$

Definition 4 (Binary Union).

$$A \cup B = \bigcup\{A, B\}$$

Definition 5 (Intersection).

$$I = \bigcap B \Leftrightarrow B \neq \varnothing \land \forall c\,(c \in I \Leftrightarrow \forall P \in B\,(c \in P))$$

Definition 6 (Binary Intersection).

$$A \cap B = \bigcap\{A, B\}$$

Axiom 5 (Power Set).

$$\forall X,\, \exists P,\, \forall S\,(S \in P \Leftrightarrow \forall s \in S\,(s \in X))$$

Definition 7 (Power Set).

$$P = \mathcal{P}(X) \Leftrightarrow \forall S\,(S \in P \Leftrightarrow \forall s \in S\,(s \in X))$$

Definition 8 (Subset).

$$S \subseteq X \Leftrightarrow S \in \mathcal{P}(X)$$

Definition 9 (Successor).

$$\text{succ}(x) = x \cup \{x\}$$

Definition 10 (Inductive Set).

$$\text{Ind}(I) \Leftrightarrow \varnothing \in I \land \forall i \in I\,(\text{succ}(i) \in I)$$

Axiom 6 (Infinity).

$$\exists I\,(\text{Ind}(I))$$

Definition 11 (Set of Natural Numbers).

$$N = \mathbb{N} \Leftrightarrow \forall n\,(n \in N \Leftrightarrow \forall_{\text{Ind}} I\,(n \in I))$$

Definition 12 (Pair).

$$(a, b) = \{\{a\}, \{a, b\}\}$$

Axiom 7 (Foundation).

$$\forall X \neq \varnothing,\, \exists x \in X\, (x \cap X = \varnothing)$$

Definition 13 (Set of Free Variables).

$$\begin{aligned}
\text{free}(a \in b) &= \{a, b\} \\
\text{free}(\neg p) &= \text{free}(p) \\
\text{free}(p \wedge q) &= \text{free}(p) \cup \text{free}(q) \\
\text{free}(\exists x\, (p)) &= \text{free}(p) \setminus \{x\}
\end{aligned}$$

Axiom 8 (Replacement). Let $B \notin \text{free}(p)$.

$$\forall A\, (\forall x \in A,\, \exists! y\, (p) \Rightarrow \exists B,\, \forall y\, (y \in B \Leftrightarrow \exists x \in A\, (p)))$$

Definition 14 (Set Builder). Let $X \notin \text{free}(p)$.

$$X = \{a \in A \mid p\} \Leftrightarrow \forall x\, (x \in X \Leftrightarrow (x \in A \wedge p))$$

Definition 15 (Cartesian Product).

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

Definition 16 (Set of Functions).

$$Y^X = \{f \subseteq X \times Y \mid \forall x \in X, \exists! y \in Y\, ((x, y) \in f)\}$$

Definition 17 (Function).

$$\begin{aligned}
f : X \to Y &\Leftrightarrow f \in Y^X \\
f(x) = y &\Leftrightarrow (x, y) \in f
\end{aligned}$$

Definition 18 (Indexed Product).

$$\prod B = \left\{c : B \to \bigcup B \mid \forall P \in B\, (c(P) \in P)\right\}$$

Axiom 9 (Choice).

$$\forall O \left(\forall o \in O\, (o \neq \varnothing) \Rightarrow \prod O \neq \varnothing\right)$$

Definition 19 (Set Exclusion).

$$A \setminus B = \{x \in A \mid x \notin B\}$$

Definition 20 (Order of Natural Numbers)**.** Let $a, b \in \mathbb{N}$.

$$\begin{aligned} a < b &\Leftrightarrow a \in b \\ a \leq b &\Leftrightarrow (a < b \vee a = b) \end{aligned}$$

Notation 13 (Numeral)**.** Let $m \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

$$m \equiv \{n \in \mathbb{N} \mid n < m\}$$

Definition 21 (Tuple)**.** Let $n \in \mathbb{N} \setminus \varnothing$.

$$(x_1, \ldots, x_n) = \begin{cases} x_1, & \text{if } n = 1 \\ (x_1, x_2), & \text{elif } n = 2 \\ ((x_1, \ldots, x_{n-1}), x_n) & \text{otherwise} \end{cases}$$